Project Overview

This project investigates how computational analysis and machine learning can be integrated with spatial and visual design workflows through a data-driven movie-analysis pipeline. By combining textual reviews, audience comments, poster imagery, and metadata collected from IMDb, YouTube, and TMDb, the project constructs a heterogeneous multimodal dataset for sentiment analysis, clustering, classification, and cross-modal interpretation.

The workflow transforms raw textual and visual information into vectorised representations using embedding models including CLIP and ResNet50. Machine learning techniques such as clustering and sentiment classification are then applied to identify latent relationships between audience perception, genre identity, and visual characteristics.

Beyond conventional data analysis, the project further explores how computational outputs can be translated into procedural spatial forms through OpenSCAD and Blender. Statistical patterns extracted from movie reviews are mapped into parametric geometries, animation systems, and cinematic spatial compositions, establishing a continuous workflow that bridges machine learning, generative design, and digital visualisation.

The project therefore demonstrates an interdisciplinary approach that connects data science, computational design, procedural modelling, and visual storytelling within a unified design pipeline.

Part 1 – Data Analysis and Machine Learning Pipeline

1.1 Web Scraping and Data Collection

This project collects and organizes data from recent high-profile movie trailers. 12 movies are selected across three genres, with four movies per genre: action films include Mission: Impossible 7, John Wick 4, Top Gun: Maverick, and Furiosa; romance films include La La Land, Past Lives, Anyone But You, and The Notebook; horror films include Hereditary, Get Out, Smile, and Talk to Me. The dataset combines YouTube comments, IMDb reviews, and TMDb posters, including approximately 3,000 YouTube comments, 720 IMDb reviews, and 144 TMDb poster images. These data are prepared for downstream sentiment analysis, genre comparison, and multimodal movie data research.

Figure 1. TMDb poster images collected across action, romance, and horror genres.

1.2 Data Collection

EN: Multiple datasets were collected from IMDb, TMDb, and YouTube.
Multiple datasets were collected from different online platforms, including IMDb, TMDb, and YouTube, in order to construct a heterogeneous and multi-modal dataset.

The YouTube dataset consists of user-generated comments associated with movie trailers. Each entry includes fields such as comment text, author, number of likes, replies, timestamps, and metadata such as genre and movie title. This dataset captures audience reactions and subjective interpretations of films.

The TMDb dataset provides structured metadata and image resources, including movie IDs, titles, genres, and multiple poster and backdrop images. Each image is linked through a URL and stored locally, enabling further visual analysis and feature extraction.

Additionally, IMDb and YouTube comments together form a large textual corpus representing diverse user opinions across different platforms. This combination ensures both data diversity and cross-platform consistency.

YouTube

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
comment_	text	raw_text	author	votes	replies	time	is_reply	is_hearted	genre	movie_title	movie_yea	tmdb_id	imdb_id	trailer_
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UgzxQErQ	the last 20	the last 20	@druxpacl	70	0	1 year ago	FALSE	FALSE	action	Mission: In	2023	575264	tt9603212	avz06F
UgzWC1Tx	Very consi	Very consi	@jamesm	1300	22	2 years ag	FALSE	FALSE	action	Mission: In	2023	575264	tt9603212	avz06F
UgwvW0k_	After I saw	After I saw	@CHESTEF	629	12	2 years ag	FALSE	FALSE	action	Mission: In	2023	575264	tt9603212	avz06F
Ugxi86PN(	ITS TIME T	ITS TIME T	@arjundas	25	0	1 year ago	FALSE	FALSE	action	Mission: In	2023	575264	tt9603212	avz06F
UgyrGIRKs	Tom Cruis	Tom Cruis	@codenar	511	13	2 years ag	FALSE	FALSE	action	Mission: In	2023	575264	tt9603212	avz06F
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Ugz0lmy5	Who's here	Who's here	@Prodrom	39	0	1 year ago	FALSE	FALSE	action	Mission: In	2023	575264	tt9603212	avz06F

TMDb

	tmdb_id	movie_title	genre	image_typ	image_idx	url	local_path
2	575264	Mission: In	action	poster	0	https://ima	data\posters\action\575264_p0.jpg
3	575264	Mission: In	action	poster	1	https://ima	data\posters\action\575264_p1.jpg
4	575264	Mission: In	action	poster	2	https://ima	data\posters\action\575264_p2.jpg
5	575264	Mission: In	action	poster	3	https://ima	data\posters\action\575264_p3.jpg
6	575264	Mission: In	action	poster	4	https://ima	data\posters\action\575264_p4.jpg
7	575264	Mission: In	action	poster	5	https://ima	data\posters\action\575264_p5.jpg
8	575264	Mission: In	action	poster	6	https://ima	data\posters\action\575264_p6.jpg
9	575264	Mission: In	action	poster	7	https://ima	data\posters\action\575264_p7.jpg
.0	575264	Mission: In	action	backdrop	0	https://ima	data\posters\action\575264_b0.jpg
.1	575264	Mission: In	action	backdrop	1	https://ima	data\posters\action\575264_b1.jpg
.2	575264	Mission: In	action	backdrop	2	https://ima	data\posters\action\575264_b2.jpg
.3	575264	Mission: In	action	backdrop	3	https://ima	data\posters\action\575264_b3.jpg
.4	603692	John Wick:	action	poster	0	https://ima	data\posters\action\603692_p0.jpg
.5	603692	John Wick:	action	poster	1	https://ima	data\posters\action\603692_p1.jpg
.6	603692	John Wick:	action	poster	2	https://ima	data\posters\action\603692_p2.jpg
.7	603692	John Wick:	action	poster	3	https://ima	data\posters\action\603692_p3.jpg
.8	603692	John Wick:	action	poster	4	https://ima	data\posters\action\603692_p4.jpg
.9	603692	John Wick:	action	poster	5	https://ima	data\posters\action\603692_p5.jpg
20	603692	John Wick:	action	poster	6	https://ima	data\posters\action\603692_p6.jpg
21	603692	John Wick:	action	poster	7	https://ima	data\posters\action\603692_p7.jpg

Figure 2. Examples of structured datasets collected from YouTube and TMDb platforms.

Files:
- imdb_reviews.csv
- tmdb_images.csv
- tmdb_metadata.csv
- youtube_comments.csv

1.3 Vectorisation

EN: Text and image data were transformed into embeddings.

Text and image data were transformed into vector representations to enable computational analysis and cross-modal comparison.

For image data, two different vectorisation methods were applied: CLIP and ResNet50. CLIP embeddings capture semantic relationships by aligning visual features with textual concepts, making them particularly suitable for cross-modal analysis between movie posters and reviews. In contrast, ResNet50 focuses on low-level visual features such as texture and shape, providing a more traditional image representation.

A comparison between the two methods shows that CLIP produces more coherent clustering in embedding space and aligns better with textual sentiment patterns, while ResNet50 captures finer visual distinctions but lacks semantic interpretability. Based on these results, CLIP embeddings were prioritised for cross-modal analysis.

Outputs:
- text_vec/
- image_vec/

1.4 Machine Learning

Machine learning techniques were applied to analyse both textual and visual datasets, including clustering and classification models.

Clustering was performed on the vectorised data to identify latent structures within both text and image domains. The results reveal that movies tend to group according to genre and sentiment patterns, indicating consistency between audience perception and visual representation.

For classification, two models were implemented. A sentiment classifier was applied to textual data to categorise reviews into positive and negative categories, while an image classifier was used to predict movie genres based on poster features. These models demonstrate how both textual and visual data can be interpreted through machine learning.

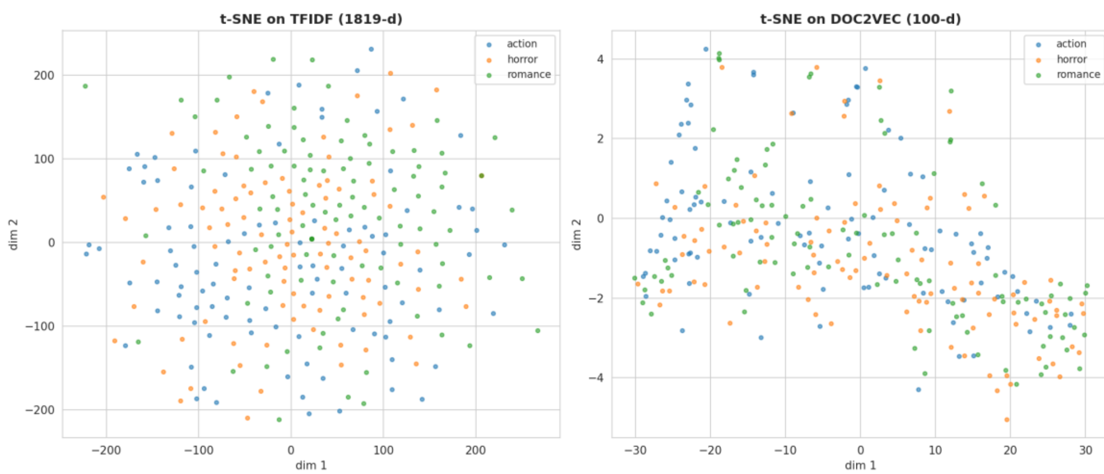


Figure 3. t-SNE projection of text embeddings showing clustering patterns across genres.

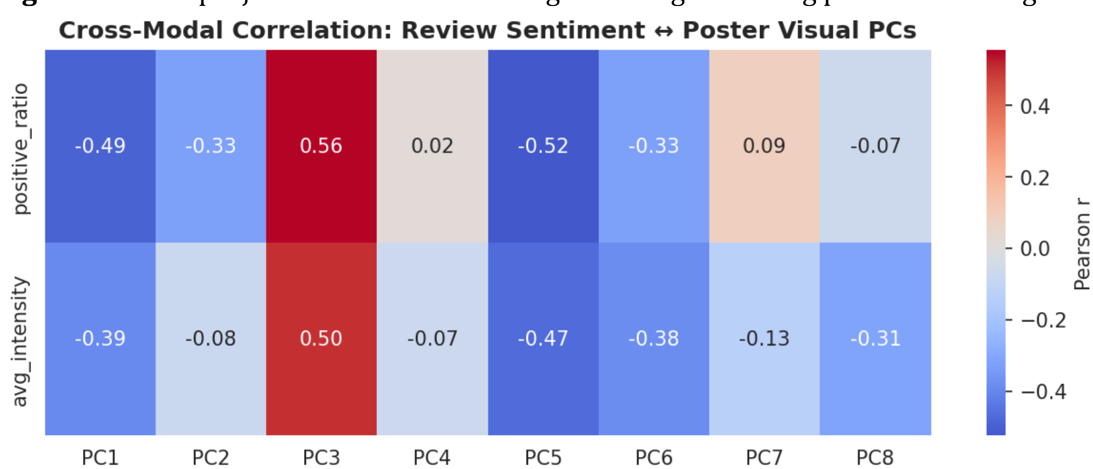


Figure 4. Cross-modal heatmap showing correlation between textual sentiment and image features.

Outputs:
- clustering/

- sentiment_clf/
- poster_clf/
- cluster_summary.csv

1.5 Visualisation

A series of visualisations were developed to analyse both textual and visual datasets and to support the interpretation of machine learning results.

The review count and sentiment distribution plots provide an overview of dataset composition and confirm a balanced representation across genres and sentiment categories. Word clouds further highlight dominant linguistic patterns, revealing differences in tone and vocabulary between genres.

To explore high-dimensional representations, t-SNE projections were applied to both text and image embeddings. These visualisations reveal clear clustering structures, indicating that the vectorisation methods successfully capture semantic and visual similarities.

Finally, the cross-modal heatmap demonstrates correlations between textual sentiment and image features, providing insight into how audience perception aligns with visual identity. Together, these visualisations bridge quantitative analysis and qualitative interpretation, forming a critical component of the data-driven workflow.

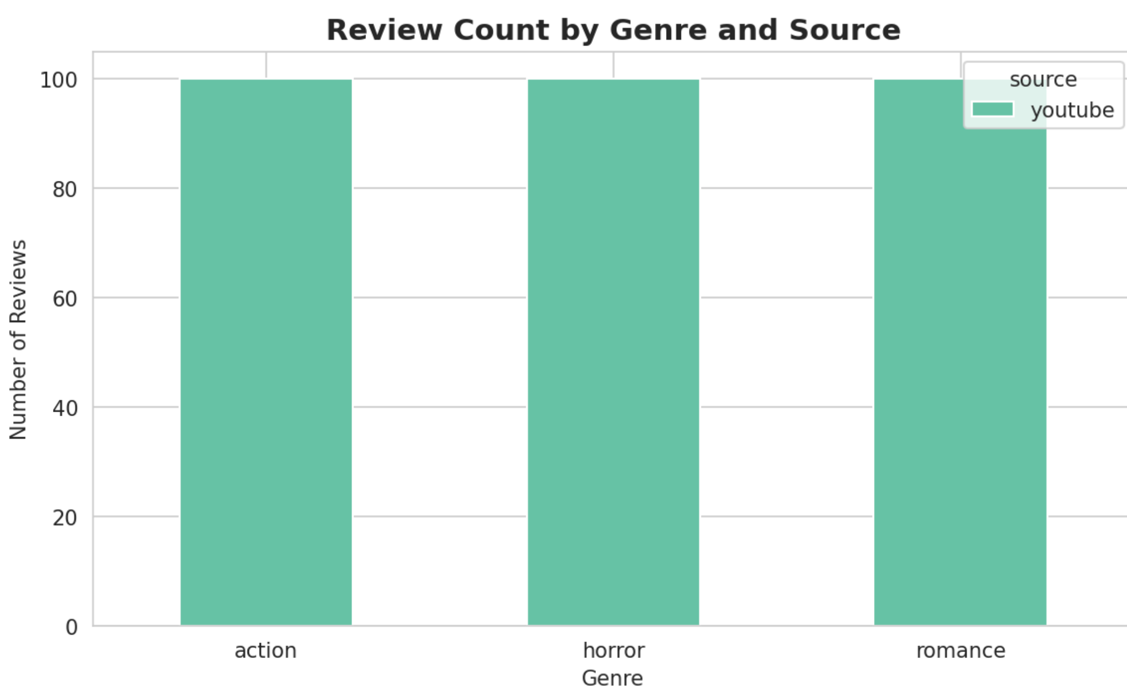


Figure 5. Review count distribution across genres.

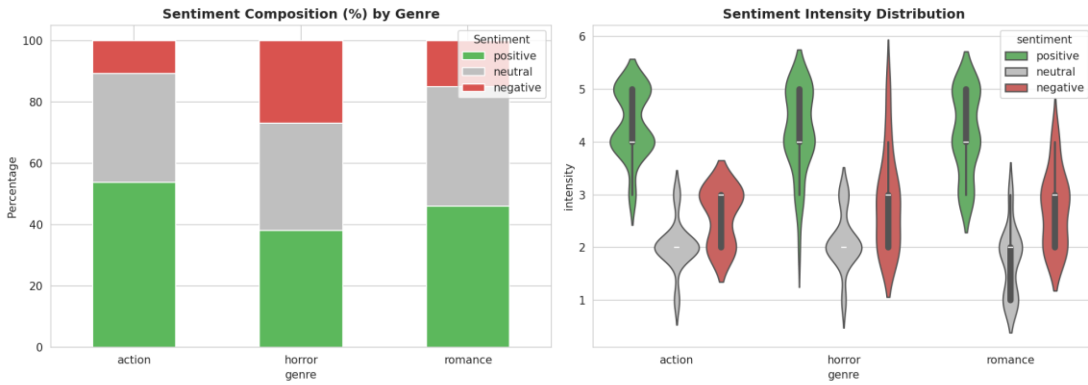


Figure 6. Sentiment distribution of movie reviews.

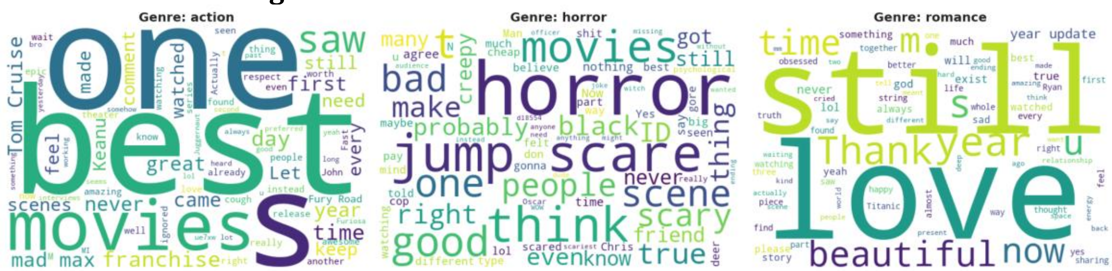


Figure 7. Word clouds showing dominant terms.

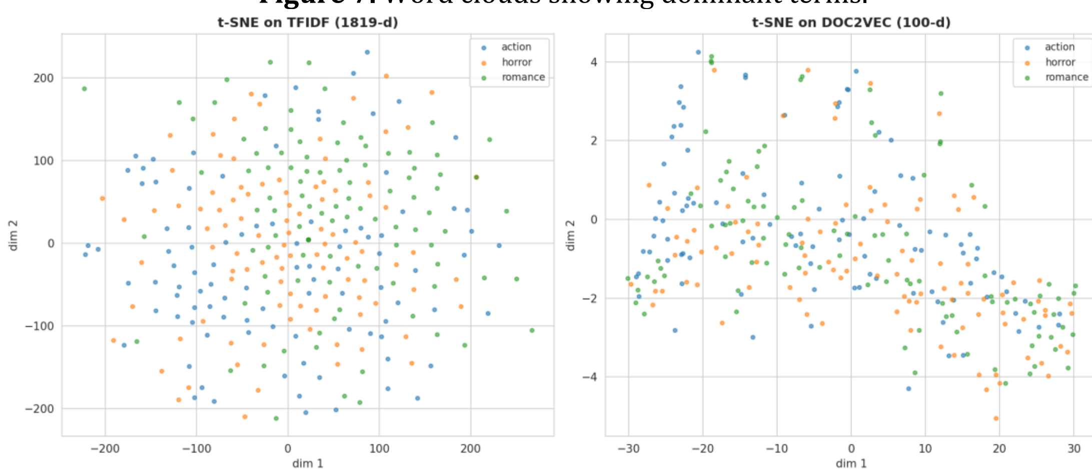


Figure 8. t-SNE projection of text embeddings.

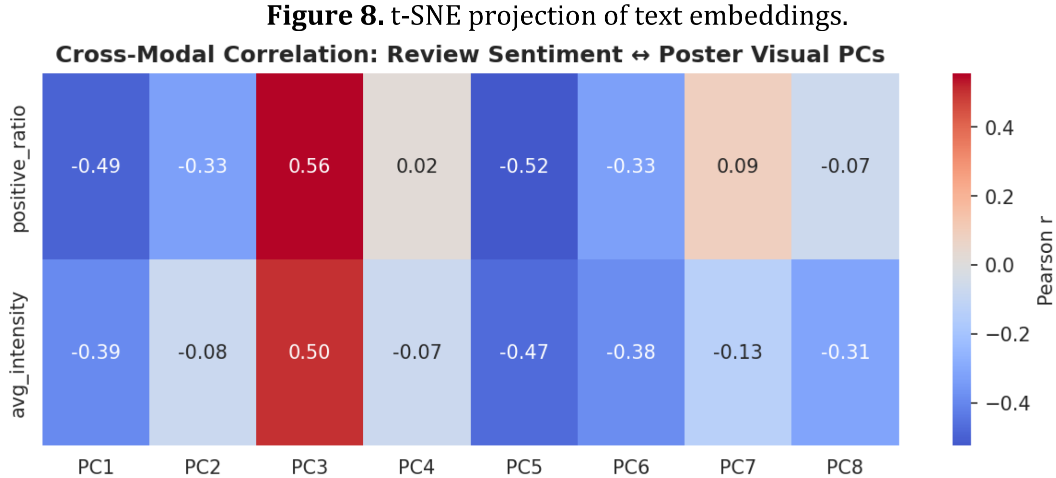


Figure 9. Cross-modal correlation heatmap.

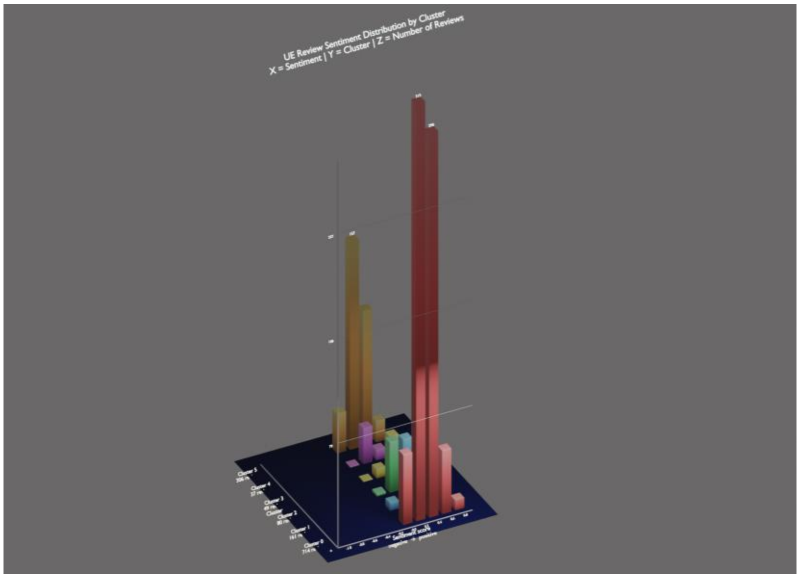


Figure 10. Data-driven 3D visualisation in Blender.

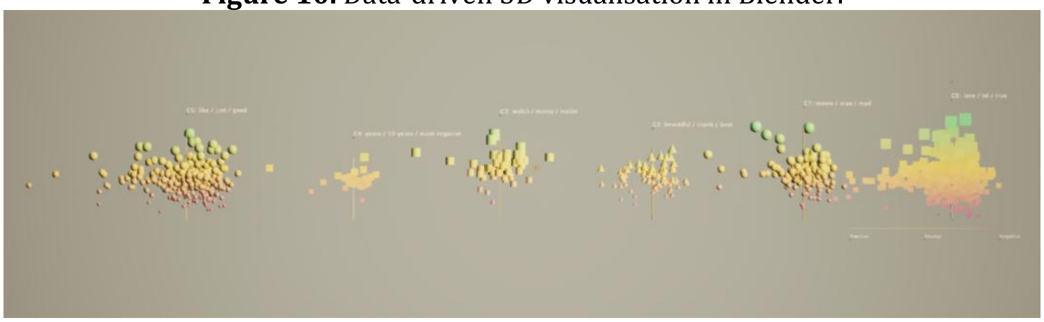


Figure 11. Spatial clustering representation in Unreal Engine.

1.6 API Integration

The GPT API was integrated into the pipeline to perform both sentiment annotation and data augmentation. During the annotation stage, the API was used to analyse movie reviews and assign sentiment labels, complementing the machine learning classification model.

In the augmentation stage, the API was used to generate additional review variations based on existing data. This process introduced stylistic diversity and expanded the dataset, improving robustness for downstream analysis.

An example of GPT-based sentiment annotation is shown below, where each review is automatically labelled using a generative API.

	A	B
1	review_text	sentiment
2	This movie was absolutely amazing, I loved every moment of it.	positive
3	The film was too long and quite boring at times.	negative
4	The acting was decent but the story was predictable.	neutral
5	Incredible visuals and great soundtrack made it very enjoyable.	positive
6	I didn't like the characters and the plot felt weak.	negative
7	A fun movie with some flaws but overall entertaining.	positive
8	The pacing was slow and it failed to keep my attention.	negative
9	Beautiful cinematography but the narrative was confusing.	neutral

Figure 12. Example of GPT-based sentiment annotation for movie reviews.

Part 2 – Design Integration

2.1 Overview

This stage translates the analytical outputs from Part 1 into spatial and visual forms using computational design tools. The processed movie-review dataset is integrated into OpenSCAD and Blender to construct a data-driven three-dimensional visualisation system.

The workflow transforms sentiment statistics, clustering structures, and embedding relationships into parametric geometries and animated spatial compositions. Rather than treating data purely as numerical information, this stage explores how machine learning outputs can become material, spatial, and visual artefacts.

The final result is a generative scene composed of multiple "sentiment steles", where each geometry represents a cluster of movie-review comments. Spatial arrangement, geometric deformation, surface erosion, and emission intensity are all driven directly by the analytical data generated in Part 1.

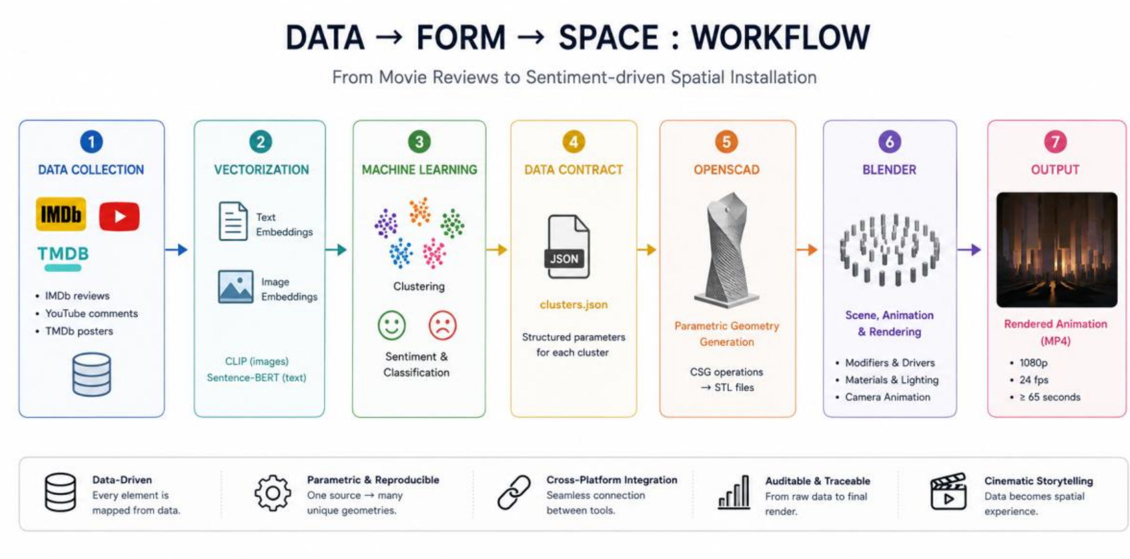


Figure 13. Computational workflow diagram connecting machine learning outputs

2.2 Design Concept

The visual concept is inspired by the idea of "review steles", where audience sentiment is transformed into monumental forms. Each stele represents one TF-IDF cluster extracted from movie reviews and trailer comments.

A total of 27 steles are distributed across three concentric rings corresponding to the three movie genres:

• Action
• Romance
• Horror

The arrangement forms a temporal "mood arc" in which geometry, scale, and illumination evolve according to emotional intensity and sentiment polarity.

The project investigates how abstract textual information can be converted into spatial experience through computational geometry and procedural animation.

2.3 OpenSCAD Parametric Geometry

OpenSCAD was used as the primary parametric modelling environment for procedural geometry generation. Unlike traditional modelling workflows, OpenSCAD operates through declarative constructive solid geometry (CSG), enabling geometries to be generated entirely through code.

Each stele is generated from a single `.scad` source file together with a set of parameters exported from the machine learning pipeline.

The following geometric operations were implemented:
• linear_extrude() for height generation
• twist deformation for controversy representation
• difference() boolean operations for erosion effects
• offset() and chamfer operations for edge shaping
• primitive combinations through union() and difference()

The generated STL geometries therefore remain fully reproducible and auditable from the source dataset.

The parametric workflow enables one base design system to produce 27 unique geometries while maintaining consistency across the overall visual language.

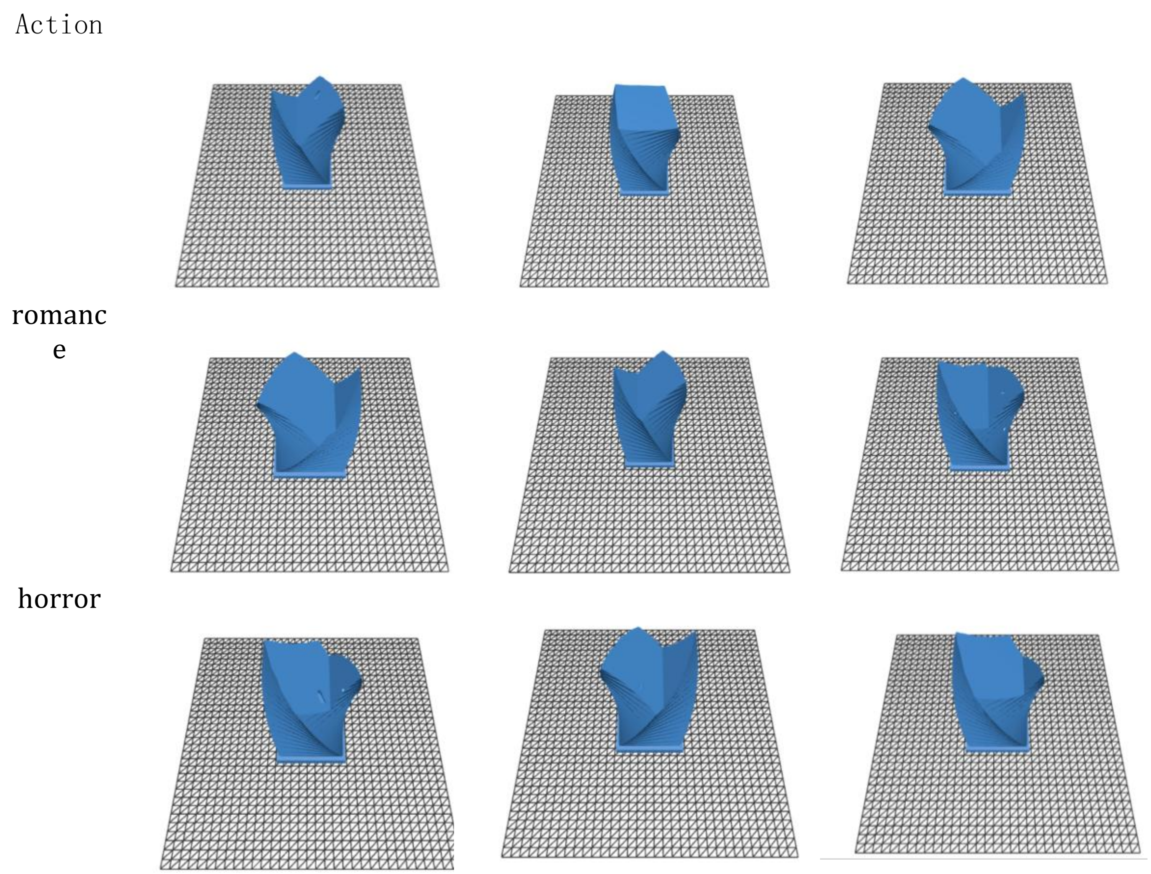


Figure14. OpenSCAD preview

2.4 Data-to-Form Mapping

A direct relationship was established between statistical outputs from Part 1 and geometric or material attributes in Part 2. This creates a transparent and auditable connection between machine learning analysis and design outcomes.

The mapping system includes:

- Average sentiment intensity → Stele height
- Cluster ID → Degree of torsion
- Vocabulary diversity → Number of helical openings
- Review count → Base width
- Sentiment polarity → Chamfer angle
- Sentiment variance → Surface erosion and emission intensity
- Poster colour statistics → Material colour

For example, clusters with higher emotional intensity generate taller geometries, while clusters with stronger controversy produce greater rotational deformation. Negative sentiment introduces erosion-like surface cavities, while positive sentiment increases emissive material properties.

This mapping process transforms statistical relationships into visually interpretable spatial structures.


```json
{
  "fragment_id": "stele-romance-11036-c0-r1",
  "genre": "romance",
  "movie_title": "The Notebook",
  "tmdb_id": 11036,
  "cluster_id": 0,
```


Figure 15. Example cluster parameter exported from the machine learning pipeline.

2.5 Blender Scene Construction

Blender 4.x was used to assemble the generated STL geometries into a complete animated scene. The software was chosen because of its flexible Python API, modifier stack system, and animation workflow.

The Blender pipeline includes:
1. STL import automation
2. Modifier stack configuration
3. Driver-based parameter binding
4. Camera-path animation
5. Material and lighting setup
6. Final rendering using EEVEE Next

Unlike geometry-node-based workflows, this project relies primarily on:

• Python scripting (bpy)
• Modifier stacks
• F-Curve drivers
• NLA animation editing

This approach keeps the workflow modular and highly controllable while maintaining procedural behaviour.

2.6 Animation System

Animation was used to visualise emotional progression across the dataset.
A central “Mood Driver” object controls global animation parameters throughout the timeline. Through Blender drivers and NLA actions, the scene gradually evolves over approximately 65 seconds.

The animation includes:
• scale variation
• emission pulsing
• camera movement
• spatial transitions
• lighting modulation

The camera follows a Bezier path around the concentric stele arrangement, allowing viewers to observe changes in geometry and lighting from multiple perspectives.
This temporal structure creates a cinematic interpretation of audience sentiment progression.

2.7 Cross-Platform Workflow

One important aspect of the project is the interoperability between analytical and design environments.
The workflow establishes a continuous pipeline:
Machine Learning
→ JSON data contract
→ OpenSCAD geometry generation
→ Blender animation
→ Rendered visual output

The shared `clusters.json` file acts as a central data contract linking all stages of the workflow.
This ensures that every visual element can be traced back to measurable analytical data, supporting reproducibility and transparency.

2.8 Design Outcome

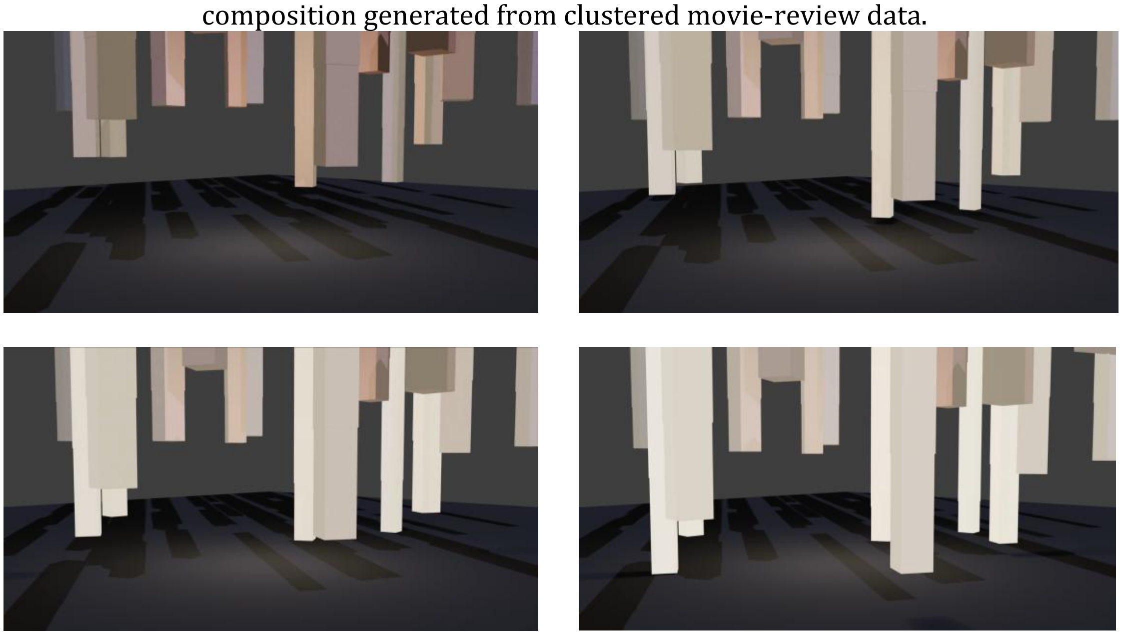
The final output consists of:
• 27 parametric STL geometries
• a complete Blender scene
• a 65-second animated visualisation
• rendered video outputs
• data-linked procedural assets

The resulting visualisation demonstrates how machine learning outputs can inform generative spatial design. Rather than functioning solely as analytical graphics, the project produces immersive visual artefacts that spatialise audience sentiment and cinematic identity.

This stage therefore bridges:
• data science
• machine learning
• procedural modelling
• animation
• spatial visualisation

within a unified computational design workflow.

Figure 16. Final rendered animation frame showing sentiment-driven spatial composition generated from clustered movie-review data.



References and Acknowledgements

References and Acknowledgements

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https://openscad.org/documentation.html

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https://pytorch.org/

NumPy documentation.

https://numpy.org/

Matplotlib documentation.

https://matplotlib.org/

OpenCV documentation.

https://opencv.org/

ResNet50 pretrained model.

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OpenCLIP.

https://github.com/mlfoundations/open_clip

This project also used OpenAI GPT models for sentiment annotation,

dataset augmentation, workflow debugging assistance, and technical writing support.